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CSE-4355 Electromechanical Systems and Sensors

THE UNIVERSITY OF TEXAS AT ARLINGTON



LAB 6 REPORT

Submitted toward the partial completion of the requirements for CSE 4355

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Introduction

In this lab we were introduced to a thermocouple, also known as a "thermoelectrical thermometer". It is an electrical device consisting of two dissimilar electrical conductors forming an electrical junction. It relies on the Sievert effect and produces a temperature-dependent voltage. Using an analog to digital converter, this voltage can be interpreted with the red board to measure temperature. We will be using this to attempt to measure the temperature of dry ice (-78.5C), ice water (0C), and a soldering iron at 300C.

Methods

First, we used a type K thermocouple jack (yellow) to connect the two output wires of the thermocouple to the two input channels of an ADS1115 A/D converter for differential measurement. To bias the inputs, we connected the positive input to Vdd through a 1 MΩ resistor and the negative input to GND through another 1 MΩ resistor. We then inserted the thermocouple into the jack. Next, we connected a TMP36 temperature sensor to one of the remaining input channels on the ADS1115. We ensured that the TMP36 was in direct contact with the thermocouple jack, as it was used to measure the cold junction temperature of the thermocouple system. Any temperature differences between the thermocouple wires in the jack and the TMP36, as well as any inaccuracies in the TMP36 sensor, could result in increased measurement error. Afterwards, we connected the ADS1115 to the TM4C123GXL development board, making sure to pull up the SDA and SCL lines with 2kΩ resistors. We then used the I2C tool to verify that the ADS1115 was properly detected.

Then to summarize the conversion, we first wrote a lookup table based on NIST ITS-90 tables which converted voltage to temperature and vice versa. We then took the raw voltage value of the TMP sensor or cold junction (Tcj) and converted it to a temperature value in C. From the data sheet we used the formula: (Tcj-0.5)/10^-3. Using the NIST table we converted this temperature back to voltage (Vcj) because we need it normalized. After this we got the raw voltage value of the thermocouple (Vtc) and added Vcj to adjust. Then with this value we put it back into the NIST table to get our final output temperature.

Initial Experiments

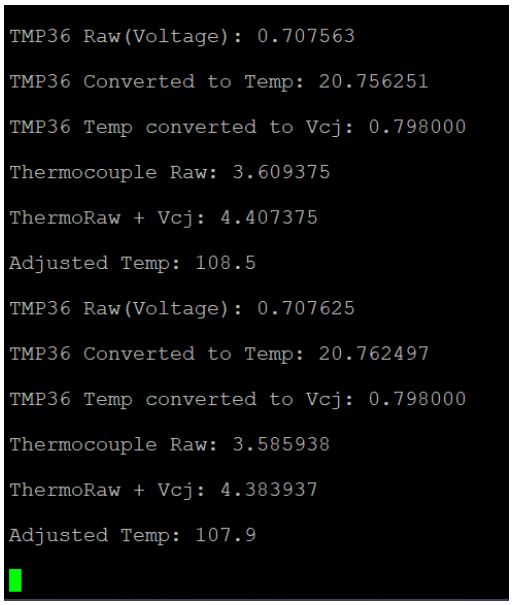
Ambient Temperature:

```
TMP36 Converted to Temp: 20.881248
TMP36 Temp converted to Vcj: 0.798000
Thermocouple Raw: 0.046875
ThermoRaw + Vcj: 0.844875
Adjusted Temp: 22.1
TMP36 Raw(Voltage): 0.708750
TMP36 Converted to Temp: 20.875001
TMP36 Temp converted to Vcj: 0.798000
Thermocouple Raw: 0.046875
ThermoRaw + Vcj: 0.844875
Adjusted Temp: 22.1
```

Ice water:

```
TMP36 Raw(Voltage): 0.707563
TMP36 Converted to Temp: 20.756251
TMP36 Temp converted to Vcj: 0.798000
Thermocouple Raw: -0.750000
ThermoRaw + Vcj: 0.048000
Adjusted Temp: 2.2
TMP36 Raw(Voltage): 0.707625
TMP36 Converted to Temp: 20.762497
TMP36 Temp converted to Vcj: 0.798000
Thermocouple Raw: -0.750000
ThermoRaw + Vcj: 0.048000
Adjusted Temp: 2.2
```

Soldering Iron:



Disconnected:

```
TMP36 Temp converted to Vcj: 0.838000

Thermocouple Raw: 255.992188

ThermoRaw + Vcj: 256.830200

Disconnected
TMP36 Raw(Voltage): 0.714250

TMP36 Converted to Temp: 21.425003

TMP36 Temp converted to Vcj: 0.838000

Thermocouple Raw: 255.992188

ThermoRaw + Vcj: 256.830200

Disconnected
TMP36 Raw(Voltage): 0.714250

TMP36 Converted to Temp: 21.425003

TMP36 Temp converted to Vcj: 0.838000

Thermocouple Raw: 255.992188

ThermoRaw + Vcj: 256.830200

Disconnected
TMP36 Raw(Voltage): 0.713938

TMP36 Converted to Temp: 21.393752

TMP36 Temp converted to Vcj: 0.838000

Thermocouple Raw: 255.992188

ThermoRaw + Vcj: 256.830200

Disconnected
```

Conclusions

In conclusion, temperature sensors and thermocouples are vital for accurate temperature measurement, compensation, and control across a wide range of applications. Thermocouples are commonly used for temperature sensing due to their ability to operate over a wide temperature range from extreme cold to extreme heat with relatively quick response times. Their integration with modern control systems and ease of use makes them indispensable tools in scientific, industrial, and commercial fields.

Circuit Built:

